Daughter spacetimes from regular black holes in higher-dimensional gravity: global structure, collapse beyond idealisation, collider production and tests

Project Baby Universe — Genesis Branch

30 October 2026

Abstract

We study a single, consistent higher-dimensional theory in which ordinary matter can produce a spacetime region that is causally disconnected from its parent: $D = 4 + n$ dimensional Einstein gravity with large extra dimensions (fundamental scale M_D) supplemented by the infinite tower of quasi-topological curvature corrections of Bueno, Cano and Hennigar, which removes the black-hole singularity. (i) The collapse of a dust ball is glued exactly to the regular static exterior; its maximal extension is curvature-bounded (with a mass-independent bound $K\,l^4 \simeq 188$ in $D = 10$), regular at both horizons and geodesically complete, and contains a white-hole region and a new asymptotically flat universe outside $J^-(\mathcal{I}^+)$ of the parent, together with a finite closed daughter (the bounced interior, whose volume grows by up to 10^{115} for a stellar collapse). (ii) Every closed universe in the theory bounces at $a \geq l$ for any perfect fluid; matter interiors accelerate only for $N \simeq \frac{D-3}{D-1} \ln(a_{\max}/l)$ e-folds, so lasting inflation would need a vacuum as dense as the hole. (iii) An energy-conserving hydrodynamics code built on the generalised Misner–Sharp equations reproduces the exact bounce to 10^{-5} and shows that, with pressure, inhomogeneity and shocks, a horizon forms and the matter bounces in an untrapped core with bounded curvature; in realistic collapse the bounced core meets the envelope at the inner horizon, and in some runs enters the white-hole region through a throat. (iv) With real parton distributions, a 100 TeV collider produces such holes iff the horizon threshold lies between $\sim$ 5–9 TeV (the LHC bound, depending on the trapped fraction) and $\sim$ 50 TeV; they form a trapped surface in 10^{-28} s, evaporate in 10^{-26} s toward a zero-temperature remnant, and the trapped region never closes, so the daughter survives. Near threshold the holes emit almost nothing and the remnant carries $\sim 95\%$ of the energy. (v) The events are a multi-TeV jet recoiling against an invisible remnant, separable from GR black holes of the same threshold with ~ 3 events; the model survives all current data (the LHC multi-object black-hole searches are blind to it), and an FCC-hh monojet search would discover it up to $M_{\rm ext} \simeq 33$ TeV at 85 TeV (40 TeV at 100 TeV). All key formulas and numbers are independently re-derived (Appendix A).

1 Introduction

Whether a laboratory process could create a new, causally disconnected spacetime region is a question about two things at once: the behaviour of gravity at high curvature, and the energy at which gravity becomes strong. In general relativity (GR) the first is blocked by singularity theorems (an inflating region in an asymptotically flat spacetime must have a singularity in its past [1]) and the second by the Planck scale. We work in a framework that removes both obstacles within one action, and ask, in turn: does a daughter spacetime exist globally; does it survive realistic collapse; can a collider make it; what could be measured; does it survive current data; and how would an experiment find it.

2 Model U

2.1 Action and spherical sector

The theory is D-dimensional gravity with an infinite tower of quasi-topological densities [2]. Its spherically symmetric sector has second-order equations, a Birkhoff theorem and Israel-type junction conditions [3, 4], and is controlled by one function h through the generalised Misner–Sharp relation

$$h(\psi) = \frac{\mu}{r^{D-1}}, \qquad \psi = \frac{1-(\nabla r)^2}{r^2}, \qquad \mu = \frac{16\pi G_D M}{(D-2)\Omega_{D-2}}. \tag{1}$$

We use the Hayward tower $h = \psi/(1-\alpha\psi)$ and the Dymnikova-type tower $h = -\ln(1-\alpha\psi)/\alpha$; both bound $\psi < 1/\alpha \equiv 1/l^2$. For closed FRW regions $\psi = H^2 + 1/a^2$ and the generalised Friedmann equation reads $h(H^2+1/a^2) = 16\pi G_D\rho/((D-1)(D-2))$ [5].

2.2 Regular black holes

The static solution $f = 1 - r^2\psi$ has a de Sitter core $f \simeq 1 - r^2/l^2$ and two horizons above the extremal mass; for the Hayward tower

$$\mu_{\rm ext} = l^{D-3}\Big(\tfrac{D-1}{D-3}\Big)^{\frac{D-1}{2}}\tfrac{D-3}{2}, \qquad T = \frac{(D-3)r_+^2 - (D-1)l^2}{4\pi r_+^3}, \tag{2}$$

so evaporation ends at a zero-temperature remnant.

2.3 Large extra dimensions

With $G_D = 1/M_D^{n+2}$ (Dimopoulos–Landsberg convention [6]) and $l = c_l/M_D$, the extra dimensions have radius $R = (M_P^2/M_D^{n+2})^{1/n}/2\pi$, 10^3–10^6 times larger than $1/M_D$: collider holes are D-dimensional and asymptotically flat. The horizon threshold is $M_{\rm ext} = \kappa_n c_l^{n+1} M_D$ with $\kappa_n = 32$–56 (Hayward) or 13–17 (Dymnikova) for $n = 4$–7. Experiments quote the fundamental scale in the PDG convention [12], $G_D = (2\pi)^n/(8\pi M_D^{n+2})$; the two are related by $M_D^{\rm DL} = [8\pi/(2\pi)^n]^{1/(n+2)} M_D^{\rm PDG} = 0.89$–$0.38\, M_D^{\rm PDG}$ for $n = 2$–6. All M_D in this paper are Dimopoulos–Landsberg.

3 The global daughter spacetime

Proposition 1 (Exact gluing)**.** *A closed dust ball of mass parameter C and boundary angle χ_b glued to the static exterior with $\mu = C\sin^{D-1}\chi_b$ satisfies $f(R) + \dot R^2 = \cos^2\chi_b$: its surface is a radial geodesic of the exterior with energy $E = \cos\chi_b$.*

The maximal extension of the exterior is tiled by patches I (parent) → II (black hole) → III (de Sitter core) → IV (white hole) → I′ (new universe), matched through the finite null coordinate (Fig. 1).

Proposition 2 (Global properties)**.** *(a) $\max K\, l^4$ is independent of the mass over $\mu = 10^{-3}$–10^8: 29–34 ($D = 4$), 45–48 ($D = 5$), 188 (Hayward, $D = 10$), 919 (Dymnikova, $D = 10$); the closed interior peaks at the de Sitter value $2D(D-1)/l^4$. (b) At both horizons the Kruskal metric function has equal finite non-zero limits from both sides. (c) Radial and non-radial, timelike and null geodesics cross every region in finite affine parameter and never terminate. (d) For $\mu > \mu_{\rm ext}$ the surface reaches I′ in finite proper time (2.5–$35\, l$); for $\mu < \mu_{\rm ext}$ it bounces back into the parent.*

The region IV ∪ I′ lies outside $J^-(\mathcal{I}^+)$ of the parent, has its own null infinity and ADM mass μ; the bounced closed interior is a finite daughter whose volume grows by $(a_{\max}/l)^{D-1}$: 1.3×10^5 ($C = 10^4$, $D = 10$) to 8×10^{114} for a four-dimensional stellar collapse.

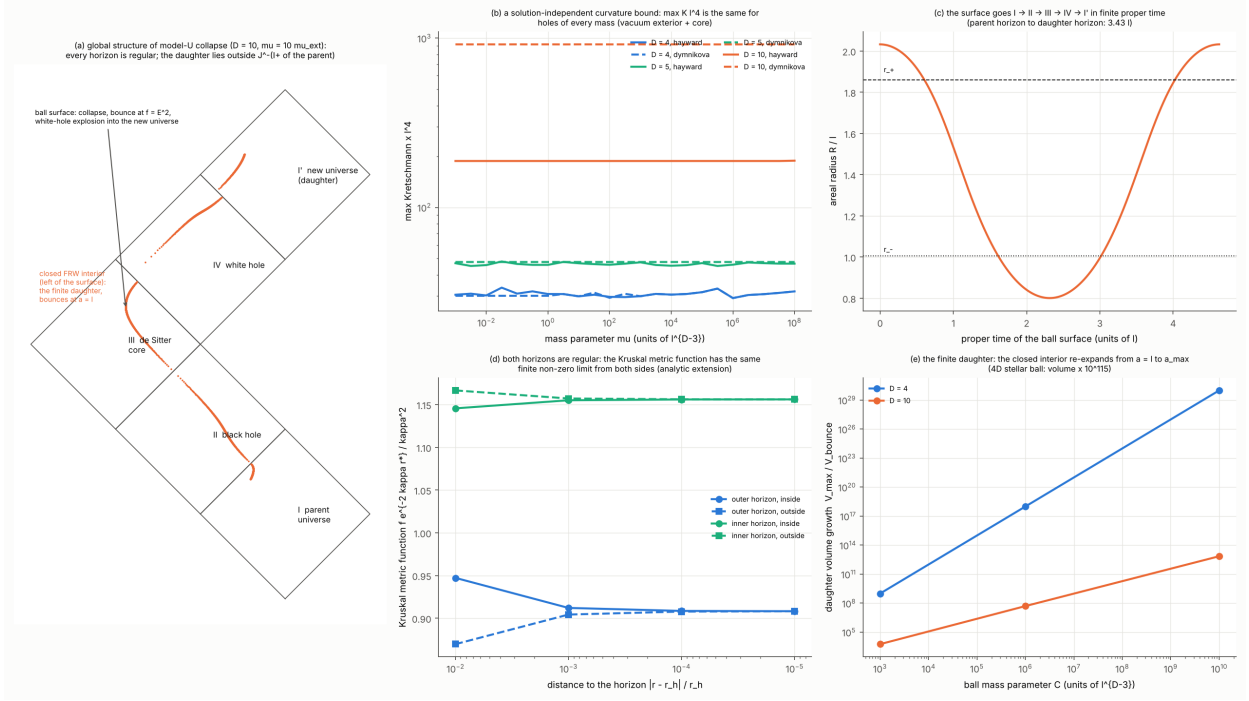


Figure 1: Global structure (Round 6m): (a) Penrose patches with the ball surface; (b) mass-independent curvature bound; (c) radius history; (d) Kruskal regularity at both horizons; (e) the finite daughter's volume growth.

4 Interiors

Proposition 3 (Universal bounce)**.** *In a closed FRW region with any perfect fluid,* $\dot{a}^2 = a^2\psi - 1 < 0$ *for* $a < l$*, so every closed universe bounces at* $a_b \geq l$*.*

For dust the turning points coincide with the static horizon radii; the interior accelerates for $N = \ln(a_{\rm infl}/a_b)$ e-folds with $a_{\rm infl}^{D-1} = 2\alpha C/(D-3)$ (Hayward), i.e. $N \to \frac{D-3}{D-1}\ln\frac{a_{\max}}{l} + \frac{1}{D-1}\ln\frac{2}{D-3}$ (29.6 for a stellar hole), then decelerates and recollapses. With a vacuum term Λ it inflates for ever iff $\Lambda > \Lambda_c(C)$, $\Lambda_c = [2(\frac{D-3}{D-1})^{(D-1)/2}/((D-3)C)]^{2/(D-3)}$ in the low-curvature limit.

Lemma 4 (Untrapped bounce)**.** *At a bounce of a radial timelike worldline,* $(\nabla R)^2 = (n\cdot\nabla R)^2 \geq 0$*; a bounce inside a black hole requires an untrapped core bounded by an inner apparent horizon. (Consequently a homogeneous Einstein–Cartan ball, whose Misner–Sharp mass drops by* 99% *at the bounce, cannot sit in a vacuum exterior.)*

5 Beyond idealised collapse

In comoving coordinates with $U = D_\tau R$ and $\Gamma = R' e^{-\lambda}$, the Lovelock structure of the spherical sector gives $\Gamma^2 - U^2 = 1 - R^2\psi(\mu/R^{D-1})$, $\mu' = (D-1)\rho R^{D-2}R'$, $D_\tau\mu = -(D-1)pR^{D-2}U$, $D_\tau\Gamma = -U\Gamma p'/((\rho+p)R')$ and

$$D_\tau U = -\frac{\Gamma^2 p'}{(\rho+p)R'} + R\psi - \frac{D-1}{2}R\psi'(y)(y+p), \qquad y = \mu/R^{D-1}, \tag{3}$$

which reduces to GR and to the generalised Friedmann equation. An energy-conserving Lagrangian scheme with non-homologous artificial viscosity reproduces the exact bounce to 10^{-5}. Inhomogeneous dust shells bounce at $R_b = \sqrt{k\alpha}$ (within 2%), inner shells first. With pressure, inhomogeneity and shocks (Fig. 2) every run forms a horizon and bounces in an untrapped core with $\psi \leq 0.994/\alpha$ and exact energy conservation. The bounced core meets the infalling envelope at the inner horizon;

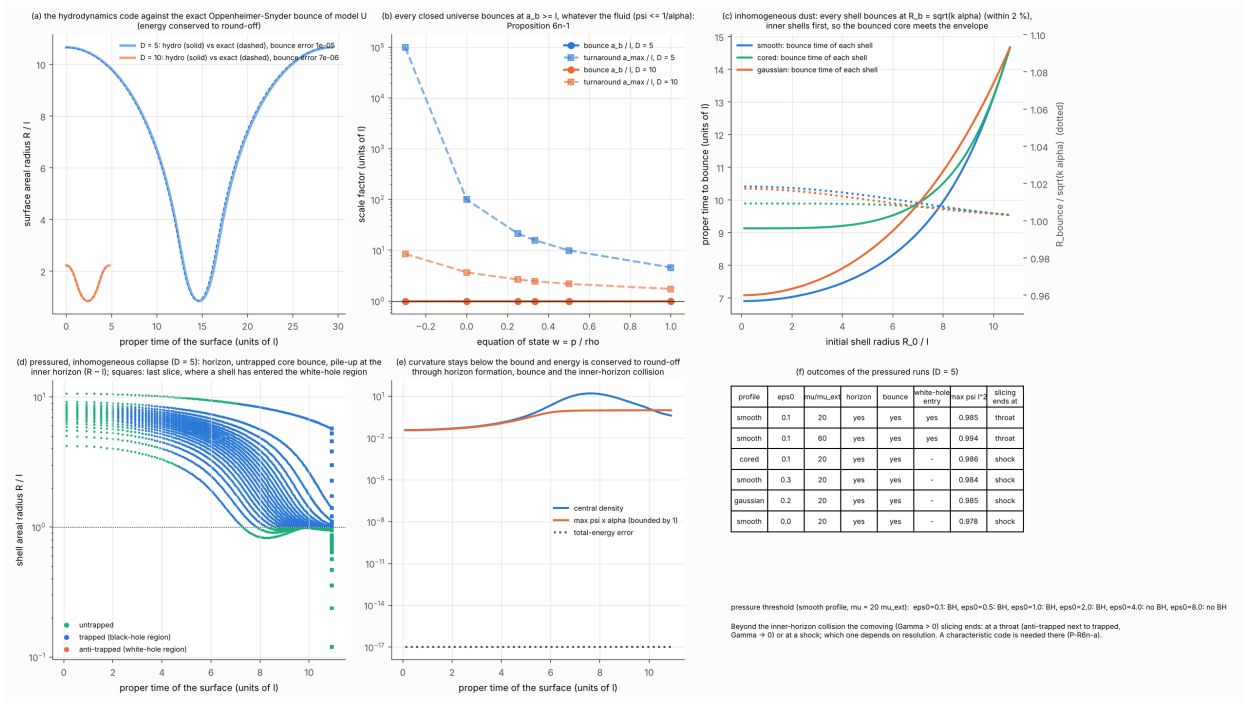


Figure 2: Beyond idealised collapse (Round 6n): (a) hydro vs exact bounce; (b) universal bounce; (c) inhomogeneous dust; (d) pressured collapse in a spacetime diagram; (e) bounded curvature and exact energy conservation; (f) outcomes.

in some runs its outer shells become anti-trapped next to trapped envelope shells with $\Gamma \to 0$ between them — a throat joining the parent's black-hole region to the daughter's white-hole region. The comoving slicing cannot cross that neck; a characteristic code is needed beyond it. Null thin shells do not exist in these theories [7], so thin-shell mass-inflation arguments do not apply; regular cores of Hayward type can be stabilised by phase-space attractors [8, 9].

6 Collider formation, evaporation and the daughter

Cross-sections use NNPDF2.3 LO [10] (momentum sum 0.998), $\hat{\sigma} = F_n \pi R_S^2$ with the Yoshino–Nambu factors [11], and the head-on trapped mass $y_n\sqrt{\hat{s}}$ ($y_0 = 1/\sqrt{2}$). Holes above $M_{\rm ext}$ are produced at 100 TeV with 30 ab^{-1} (at least 10 events) iff $M_{\rm ext} \lesssim 46$–51 TeV, and are compatible with the LHC iff $M_{\rm ext} \gtrsim 5$–9 TeV (Sec. 8): a window in c_l of width ~ 0.15–0.25. In the evaporating model-U Vaidya geometry $f(v,r) = 1 - r^2\psi(\mu(v)/r^{D-1})$ with brane Stefan–Boltzmann emission [18], $\mu - \mu_{\rm ext} \propto v^{-1}$ and $r_+ - r_- \propto v^{-1/2}$: the trapped region never closes, and the bounced partons reach the boundary of the parent's chart at finite affine parameter ($\simeq 2\,l$). Formation takes $\sim 10^{-28}$ s and evaporation 10^{-26}–10^{-25} s. An event generator built on this physics (Fig. 3) shows that near threshold the holes are nearly extremal and emit almost nothing (no Hawking quantum in 76–85% of events), the remnant carrying $\simeq 95\%$ of the mass.

7 Distinguishable predictions

We pass generator events through a lab frame (hole rapidity from the parton momentum fractions, leading-log initial-state radiation $P(p_T > p) = 1 - \exp[-A \ln^2(\sqrt{\hat{s}}/p)]$, $A \simeq 0.043$) and a generic detector ($|\eta| < 4$, $p_T > 50$ GeV; neutrinos and a neutral remnant invisible). At 100 TeV ($n = 6$, $M_D = 8$ TeV, $c_l = 0.6$, $M_{\rm ext} = 11.5$ TeV) model-U events have on average one visible object; 56–63% are a single initial-state jet balanced by the invisible remnant and 20–22% have no visible object, with $\langle {\rm MET}/S_T \rangle = 0.94$–$0.97$. A GR hole of the same threshold and cross-section gives

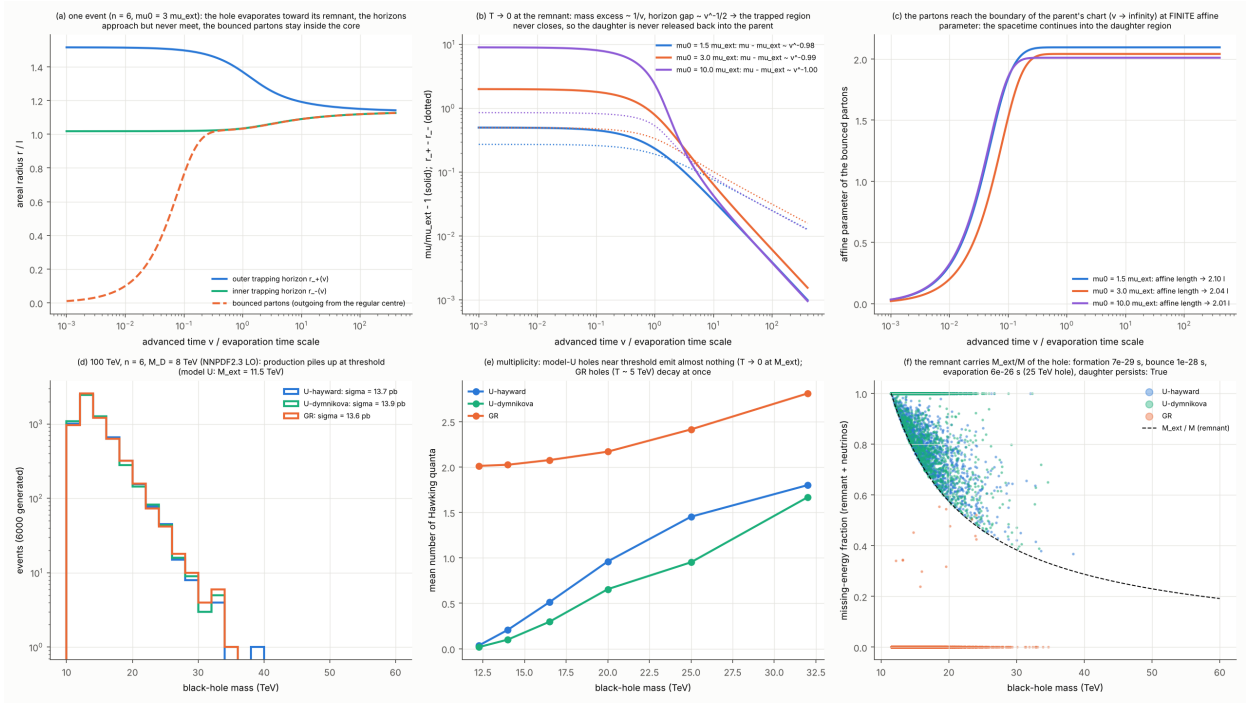


Figure 3: One event and its observables (Round 6o): (a)–(c) evaporation toward the remnant, the horizons and the bounced partons; (d)–(f) generator output at 100 TeV.

2.8 objects, median $S_T = 12\,\text{TeV}$ and $\text{MET}/S_T = 0.18$. The expected log-likelihood ratio of the binned (multiplicity, MET/S_T) distributions separates model U from GR at 5σ with ~ 3 events, and the Hayward from the Dymnikova tower with ~ 660. The four distinguishing predictions are the onset at $M_{\text{ext}} = \kappa_n c_l^{n+1} M_D$ rather than a few M_D; the invisible-remnant monojet topology; a temperature that rises from zero at M_{ext}; and heavy, slow ($\langle\beta\rangle \simeq 0.39$) remnants (Fig. 4). Kaluza–Klein graviton emission and species democracy are shared by all interior classes and test only low-scale gravity and the black-hole nature of the events.

8 Confrontation with data

Convention. Earlier versions of this analysis compared M_D^{DL} directly with experimental M_D^{PDG}; converted, the CMS monojet bounds $M_D^{\text{PDG}} > 9.9, 7.5, 6.3, 5.7, 5.3\,\text{TeV}$ ($n = 2$–6) [13] become $M_D^{\text{DL}} > 8.8, 4.7, 3.2, 2.4, 2.0\,\text{TeV}$. All benchmarks used here are allowed. *LHC.* Near-threshold model-U events populate the monojet high-recoil region; the ATLAS model-independent limit $\sigma \times A \times \epsilon < 0.3\,\text{fb}$ [14] (efficiency 0.5 assumed) excludes $M_{\text{ext}} \lesssim 8$–$9\,\text{TeV}$ with full trapping, or $\lesssim 4.7$–$5.5\,\text{TeV}$ with the head-on trapped fraction ($n = 4$–7). Multi-object black-hole searches (high S_T, ≥ 3 objects) accept only 1–6% of model-U events (75–97% for GR holes); their limits do not apply. *Neutron stars.* A remnant captured in a neutron star accretes $\sim 10^{13}$–$10^{14}\,\text{GeV/s}$ and re-radiates it from $(2\text{–}40) \times 10^{-9}$ above extremality at $T \simeq 0.16\,\text{GeV}$: there is no growth, consistent with the survival of neutron stars (cf. [15]). *Future machines.* With at least 10 holes (head-on trapping), the reach in M_{ext} is 6.3 TeV at the HL-LHC, 12.7 TeV at 27 TeV, 40.6 TeV at FCC-hh (85 TeV, $20\,\text{ab}^{-1}$) and 48.2 TeV at 100 TeV ($30\,\text{ab}^{-1}$).

9 Experimental design

Selection. Leading object $p_T > p_{\text{cut}}$, $\text{MET} > p_{\text{cut}}$ and $\text{MET}/S_T > 0.8$, with p_{cut} optimised between 3 and 16 TeV. *Background.* $Z(\to \nu\bar{\nu})$+jet with $\text{MET} > p_{\text{cut}}$, multiplied by 3 for $W(\to \ell\nu)$+jet with a lost lepton (order of magnitude). *Statistics.* Discovery requires the Asimov significance

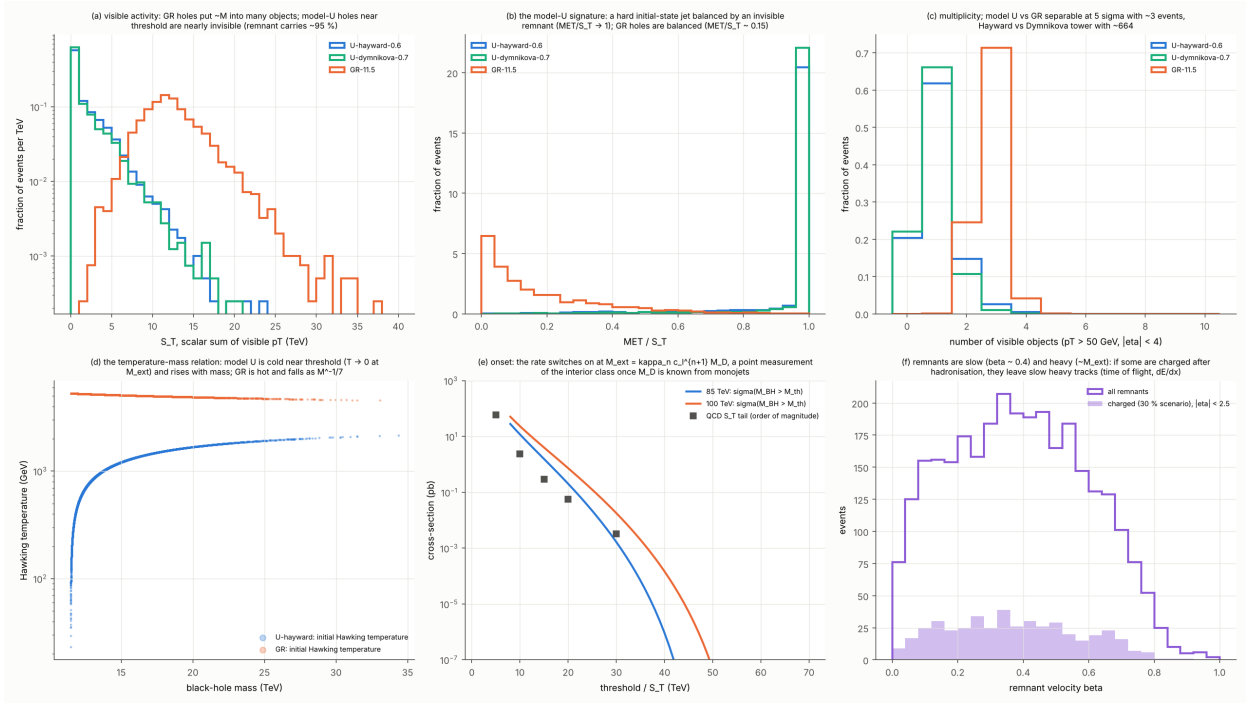


Figure 4: Model U against a GR hole at the same threshold (Round 7): (a) visible activity; (b) MET/S_T; (c) multiplicity and the number of events needed to separate the classes; (d) the temperature–mass relation; (e) the onset; (f) remnant velocities.

[16] $Z = \sqrt{2[(s+b)\ln(1+s/b) - s]} = 5$ and $s \geq 5$. For $n = 6$, $M_D = 8\,\text{TeV}$ the 5σ reach is $M_{\text{ext}} \simeq 33.5\,\text{TeV}$ at FCC-hh (85 TeV, $20\,\text{ab}^{-1}$) and 40.0 TeV at 100 TeV ($30\,\text{ab}^{-1}$); ten selected events (class and onset) are obtained up to 35.8 and 44.2 TeV, and, if 30% of remnants are charged, ten heavy stable charged tracks up to 39.1 and 46.3 TeV. Both towers agree within 1 TeV. Most of the window is decided with very little luminosity ($7\,\text{pb}^{-1}$ at $M_{\text{ext}} = 11\,\text{TeV}$, $2\,\text{fb}^{-1}$ at 21 TeV). *Detector.* Unprescaled MET and jet triggers at 1 TeV, calorimetry to $|\eta| \simeq 6$ as in the FCC-hh reference design [17], 30–50 ps timing (a $\beta = 0.39$ remnant arrives 52 ns late at 10 m) and dE/dx ($\simeq 7\times$ minimum ionising), giving $M = p/(\beta\gamma)$. *Interpretation.* A KK excess without an onset up to the reach, an onset with balanced high-multiplicity events, and an onset at $M_{\text{th}} \gg M_D$ with invisible remnants respectively point to c_l above the window, a GR-like exterior (whose interior experiment cannot decide), and the model-U class, in which by the closure lemma the trapped region never closes and the daughter persists (Fig. 6).

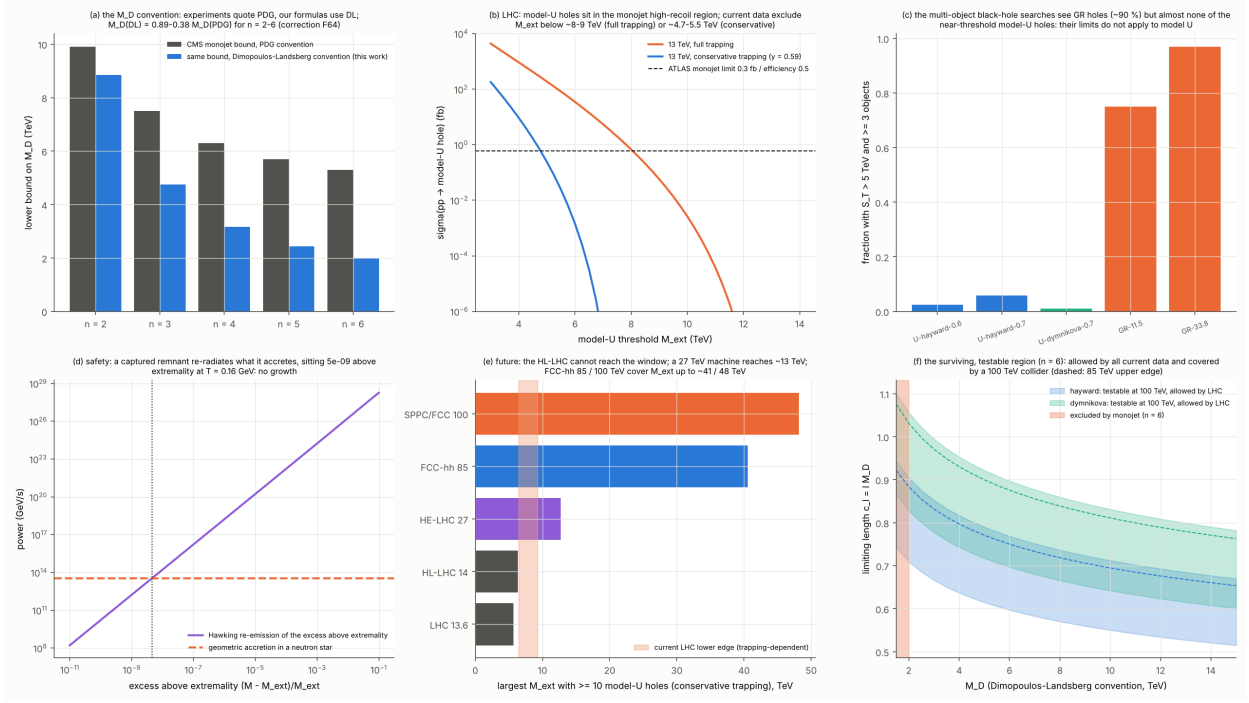


Figure 5: Model U against current and future data (Round 8): the convention correction, the LHC monojet bound, the acceptance of black-hole searches, neutron-star safety, future reach and the surviving (M_D, c_l) region.

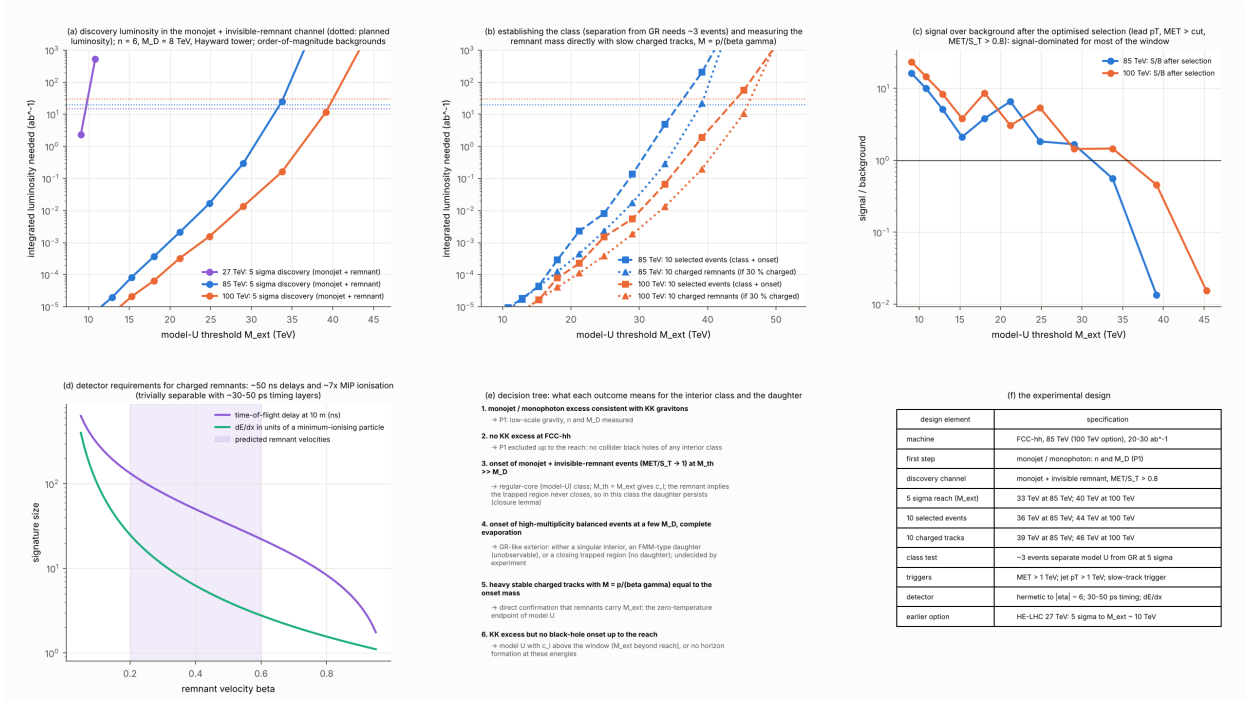


Figure 6: The experimental design (Round 9): discovery, class and remnant-mass luminosities; signal over background; detector requirements; decision tree; specification.

10 Summary of results

Quantity	Value	Status
Horizon threshold	$M_{\rm ext} = \kappa_n c_l^{n+1} M_D$, $\kappa_n = 13$–56	exact (audited)
Curvature bound ($D = 10$)	$\max K\, l^4 = 188$ (Hayward), 919 (Dymnikova), all masses	numerical (audited)
Closed-universe bounce	$a_b \geq l$ for every perfect fluid	proposition
Transient acceleration	$N \simeq \frac{D-3}{D-1} \ln(a_{\max}/l)$; 29.6 for a stellar hole	exact (audited)
Lasting inflation	needs $\Lambda > \Lambda_c(C)$ (hole-density vacuum)	exact (audited)
Finite daughter	volume growth $(a_{\max}/l)^{D-1}$, up to 8×10^{114}	exact
Hydrodynamic bounce	every run; $\psi \leq 0.994/\alpha$; energy exact; throat in 2/6	numerical
Collider window	5–$9 \lesssim M_{\rm ext} \lesssim 50\,\text{TeV}$ at 100 TeV	numerical (audited)
Evaporation	$\mu - \mu_{\rm ext} \propto v^{-1}$, gap $\propto v^{-1/2}$, never closes	numerical (audited)
Near-threshold events	0.2–0.3 quanta; remnant carries $\simeq 95\%$	Monte Carlo
Separation from GR	~ 3 events at 5σ (towers: ~ 660)	Monte Carlo
Current data	monojet, BH searches, neutron stars: consistent	numerical
FCC-hh 5σ reach	$M_{\rm ext} \simeq 33\,\text{TeV}$ (85 TeV), 40 TeV (100 TeV)	Monte Carlo, o.o.m. bkg.

Table 1: Main results. "Audited" means re-derived independently (Appendix A).

11 Limitations

Rotation is not included (collider holes have large spin; no rotating model-U solutions are known); greybody factors are omitted; backgrounds are order-of-magnitude estimates and no detector simulation (with FCC-hh pile-up) has been done; the remnant charge is unknown; the throat stage of inhomogeneous collapse needs a characteristic code; the threshold window depends on the unknown coefficient c_l; and the daughter itself is outside $J^-(\mathcal{I}^+)$, so experiments can test the interior class (onset, $T(M)$, remnants) but not the daughter directly.

A Independent audit

Each key result was re-derived by a different route (`audit6p.py`): symbolic derivations of $\mu_{\rm ext}$ and T ($D = 4, 5, 6, 10$), of $a_{\rm infl}$, of Λ_c and of the hydrodynamic acceleration equation with its GR limit; direct Riemann-tensor computations of the static and FRW Kretschmann formulas ($D = 4, 5$); the Dimopoulos–Landsberg radius from the Tangherlini mass; time integration of the interior against quadrature; closed-form curvature against finite differences; Monte Carlo cross-sections over the raw parton grid against the luminosity table; and an independent ODE integrator for the evaporation. The ledger is in `AUDIT_ROUND6P.md`.

References

[1] E. Farhi, A. H. Guth, Phys. Lett. B **183**, 149 (1987).

[2] P. Bueno, P. A. Cano, R. A. Hennigar, Phys. Lett. B **861** (2025), arXiv:2403.04827.

[3] P. Bueno, P. A. Cano, R. A. Hennigar, Á. J. Murcia, Phys. Rev. D **111**, 104009 (2025), arXiv:2412.02740.

[4] P. Bueno, P. A. Cano, R. A. Hennigar, Á. J. Murcia, Phys. Rev. Lett. **134**, 181401 (2025), arXiv:2412.02742.

[5] P. Bueno, P. A. Cano, R. A. Hennigar, Á. J. Murcia, A. Vicente-Cano, arXiv:2505.09680.

[6] S. Dimopoulos, G. Landsberg, Phys. Rev. Lett. **87**, 161602 (2001).

[7] F. Di Filippo, D. Kubizňák, A. Srinivasan, Phys. Rev. D **114**, 044061 (2026), arXiv:2604.27980.

[8] A. Bonanno, A.-P. Khosravi, F. Saueressig, arXiv:2010.04226.

[9] A. Bonanno, D. Panassiti, F. Saueressig, arXiv:2507.03581.

[10] NNPDF Collaboration, Nucl. Phys. B **867**, 244 (2013).

[11] H. Yoshino, Y. Nambu, Phys. Rev. D **67**, 024009 (2003).

[12] G. F. Giudice, R. Rattazzi, J. D. Wells, Nucl. Phys. B **544**, 3 (1999), hep-ph/9811291.

[13] CMS Collaboration, Phys. Rev. D **97**, 092005 (2018), arXiv:1712.02345.

[14] ATLAS Collaboration, Phys. Rev. D **103**, 112006 (2021), arXiv:2102.10874.

[15] S. B. Giddings, M. L. Mangano, Phys. Rev. D **78**, 035009 (2008).

[16] G. Cowan, K. Cranmer, E. Gross, O. Vitells, Eur. Phys. J. C **71**, 1554 (2011), arXiv:1007.1727.

[17] FCC Collaboration, Eur. Phys. J. Spec. Top. **228**, 755 (2019).

[18] R. Emparan, G. T. Horowitz, R. C. Myers, Phys. Rev. Lett. **85**, 499 (2000).